Math reference

Latin American regional preparation. C++17; zero-based indices; array ranges $[l, r)$ unless stated. Choose the model, check bounds, then type. Edition 2026-10-18.

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Math (1)

1.1 Exact arithmetic

Prelude.h

Description: Shared includes, aliases and bounds. Include before the modules.

<algorithm>, <cstdint>, <limits>, <map>, <numeric>, <optional>, <stdexcept>, <string>, <tuple>, <utility>, <vector> 17bffa, 8 lines

```
namespace finals_math {
using namespace std;
using i64 = long long;
using vi = vector<int>;
using vl = vector<i64>;
using i128 = __int128_t; // GCC/Clang

} // namespace finals_math
```

1.1.1 Floor, ceiling and bounds

Use: signed floor/ceiling division; multiples in $[L, R]$: $\max(0, \lfloor R/d \rfloor - \lceil L/d \rceil + 1)$, $d > 0$. **Pitfall:** C++ truncates toward zero. Widen *before* multiplying or subtracting: `i128(a)*b`. Helpers throw for zero denominator or overflow; 128 bits is finite.

IntegerDivision.h

Description: Checked narrowing and signed division.

"Prelude.h" cccfe3, 21 lines

```
namespace finals_math {
inline i64 narrow(i128 x) {
  if (x < numeric_limits<i64>::min() ||
      x > numeric_limits<i64>::max())
    throw overflow_error("signed 64-bit result");
  return static_cast<i64>(x);
}
// Throws for b=0 or quotient overflow.
inline i64 floor_div(i64 a, i64 b) {
  if (!b) throw invalid_argument("zero denominator");
  i128 x = a, y = b, q = x / y, r = x % y;
  if (r && ((r < 0) != (y < 0))) --q;
  return narrow(q);
}
inline i64 ceil_div(i64 a, i64 b) {
  if (!b) throw invalid_argument("zero denominator");
  i128 x = a, y = b, q = x / y, r = x % y;
  if (r && ((r < 0) == (y < 0))) ++q;
  return narrow(q);
}
} // namespace finals_math
```

1.1.2 Exact decimal input

Use: `parse_fixed(s,k)` stores $s10^k$ exactly; `digits10(x)` counts magnitude digits, including one for zero. **Requires:** $0 \le k \le 18$, integer part present, at most k fractional places, signed 64-bit result. No rounding/exponents/commas; read text, not `double`.

ExactDecimals.h

Description: Exact decimals and digit counts.

"Prelude.h", "IntegerDivision.h" 94a11e, 50 lines

```
namespace finals_math {
// Exact signed decimal; 0<=places<=18; excess digits rejected.
inline i64 parse_fixed(const string &s, int places) {
  if (places < 0 || places > 18)
    throw invalid_argument("decimal places");
  size_t j = 0;
  bool neg = false;
  if (j < s.size() && (s[j] == '+' || s[j] == '-'))
    neg = s[j++] == '-';
  i128 value = 0;
  int whole = 0, frac = 0;
  const i128 cap = i128(numeric_limits<i64>::max()) + 1;
  auto digit = [&](char c) {
    if (c < '0' || c > '9')
      throw invalid_argument("decimal digit");
    value = value * 10 + (c - '0');
    if (value > cap) throw overflow_error("decimal value");
  };
  while (j < s.size() && s[j] != '.') {
    digit(s[j++]);
    ++whole;
  }
  if (!whole) throw invalid_argument("integer part");
  if (j < s.size()) {
    ++j;
    while (j < s.size()) {
      digit(s[j++]);
      ++frac;
    }
  }
  if (frac > places)
    throw invalid_argument("too many fractional digits");
  while (frac++ < places) {
    value *= 10;
    if (value > cap) throw overflow_error("decimal value");
  }
  return narrow(neg ? -value : value);
}
// Magnitude digits: zero has one; sign excluded.
inline int digits10(i64 x) {
  i128 v = x;
  if (v < 0) v = -v;
  int d = 1;
  while (v >= 10) {
    v /= 10;
    ++d;
  }
  return d;
}
} // namespace finals_math
```

1.2 Congruences

1.2.1 Powers and inverses

Use: normalize, multiply and exponentiate modulo $m > 0$; $O(\log e)$ power, $e \ge 0$. $0^0 = 1$ before reduction. **Pitfall:** division requires $\gcd(a, m) = 1$; $a^{-1} = a^{p-2} \pmod p$ requires prime p, $p \nmid a$.

ModularArithmetic.h

Description: Modular arithmetic and binary powers.

"Prelude.h" 0b8b9c, 21 lines

```
namespace finals_math {
inline i64 norm(i64 a, i64 m) {
  if (m <= 0)
    throw invalid_argument("positive modulus required");
  a %= m;
  return a < 0 ? a + m : a;
}
inline i64 mul_mod(i64 a, i64 b, i64 m) {
  return i64(i128(norm(a, m)) * norm(b, m) % m);
}
inline i64 pow_mod(i64 a, uint64_t e, i64 m) {
  a = norm(a, m);
  i64 r = 1 % m;
  while (e) {
    if (e & 1) r = mul_mod(r, a, m);
    a = mul_mod(a, a, m);
    e >>= 1;
  }
  return r;
}
} // namespace finals_math
```

1.2.2 Bezout and shared-factor CRT

$ax + by = g = \gcd(a, b)$; $ax + by = c$ solvable iff $g \mid c$. All solutions: $(x_0 + tb/g, y_0 - ta/g)$; intersect bounds using floor/ceiling. For $a = b = 0$, handle $c = 0$ separately.

Use: merge $X = a \pmod m$, $X = b \pmod n$ even with shared factors. $g \mid (b - a)$ required; solve $(m/g)t = (b - a)/g \pmod{n/g}$, then $X = a + mt$, modulo $\operatorname{lcm}(m, n)$. **Requires:** positive moduli; nonnegative Euclid inputs. Incompatible CRT returns `nullopt`; overflowing lcm throws.

EuclidCRT.h

Description: Extended Euclid, inverses and CRT.

"Prelude.h", "IntegerDivision.h", "ModularArithmetic.h" 9f1514, 48 lines

```
namespace finals_math {
// Requires nonnegative a,b. Returns g,x,y with a*x+b*y=g.
inline tuple<i64, i128, i128> extended_gcd(i64 a, i64 b) {
  if (a < 0 || b < 0)
    throw invalid_argument("nonnegative gcd inputs");
  i128 r = a, s = b, x = 1, y = 0, u = 0, v = 1;
  while (s) {
    i128 q = r / s;
    i128 t = r - q * s;
    r = s;
    s = t;
    t = x - q * u;
    x = u;
    u = t;
    t = y - q * v;
    y = v;
    v = t;
  }
  return {i64(r), x, y};
```

```
}
inline optional<i64> inverse(i64 a, i64 m) {
  if (m <= 1)
    throw invalid_argument("inverse modulus must exceed one");
  auto [g, x, y] = extended_gcd(norm(a, m), m);
  (void)y;
  if (g != 1) return nullopt;
  return i64((x % m + m) % m);
}
struct Congruence {
  i64 residue, modulus;
};
// Merge x=a (mod m), x=b (mod n); nullopt = incompatible.
// Positive moduli. Throws if the merged lcm exceeds LLONG_MAX.
inline optional<Congruence> crt(Congruence a, Congruence b) {
  a.residue = norm(a.residue, a.modulus);
  b.residue = norm(b.residue, b.modulus);
  auto [g, x, y] = extended_gcd(a.modulus, b.modulus);
  (void)y;
  i128 delta = i128(b.residue) - a.residue;
  if (delta % g) return nullopt;
  i64 n = b.modulus / g;
  i128 t = (delta / g * x) % n;
  if (t < 0) t += n;
  i64 lcm = narrow(i128(a.modulus) * n);
  i128 r = (i128(a.residue) + i128(a.modulus) * t) % lcm;
  return Congruence{i64(r), lcm};
}
} // namespace finals_math
```

1.3 Prime exponents and counting values

1.3.1 Factorization and totient

For $n = \prod p^{e_p}$:

$$\tau(n) = \prod (e_p + 1), \quad \sigma(n) = \prod \sum_{j=0}^{e_p} p^j, \quad \phi(n) = n \prod_{p \mid n} (1 - 1/p).$$

Use: positive inputs; trial factorization $O(\sqrt{n})$, SPF preprocessing $O(V)$ time/space, factor $x \le V$ in $O(\log x)$. **Pitfall:** repeated arbitrary 64-bit inputs need faster factorization; $n = 1$ has empty factors. Bound divisor sums/products separately.

FactorSieve.h

Description: Trial factorization, totients and sieve.

"Prelude.h" ba3fd8, 58 lines

```
namespace finals_math {
// n>0; O(sqrt(n)).
inline vector<pair<i64, int>> factor(i64 n) {
  if (n < 1) throw invalid_argument("positive factor input");
  vector<pair<i64, int>> f;
  for (i64 p = 2; p <= n / p; p += (p == 2 ? 1 : 2))
    if (n % p == 0) {
      int e = 0;
      do {
        n /= p;
        ++e;
      } while (n % p == 0);
      f.push_back({p, e});
    }
  if (n > 1) f.push_back({n, 1});
  return f;
}
inline i64 phi(i64 n) {
  i64 ans = n;
  for (auto [p, e] : factor(n)) {
    (void)e;
    ans = ans / p * (p - 1);
  }
  return ans;
}
struct Sieve {
  vi spf, primes;
  explicit Sieve(int n) {
    if (n < 0)
      throw invalid_argument("nonnegative sieve limit");
    spf.resize(size_t(n) + 1, 0);
    for (int i = 2; i <= n; ++i) {
      if (!spf[i]) {
        spf[i] = i;
        primes.push_back(i);
      }
      for (int p : primes) {
        if (p > spf[i] || p > n / i) break;
        spf[p * i] = p;
      }
    }
  }
  vector<pair<i64, int>> factors(int x) const {
    if (x < 1 || x >= int(spf.size()))
      throw invalid_argument("sieve range");
    vector<pair<i64, int>> f;
    while (x > 1) {
      int p = spf[x], e = 0;
      do {
        x /= p;
        ++e;
      } while (x > 1 && spf[x] == p);
      f.push_back({p, e});
    }
    return f;
  }
};
} // namespace finals_math
```

1.3.2 Histogram and exact gcd classes

Positive values $\le V$: $f[x]$ = frequency, $M[d] = \sum_{k \ge 1} f[kd]$; all M cost $O(V \log V)$. For each fixed b, process divisors decreasing:

$$E_b[g] = M[g] - \sum_{\substack{h \mid b,\ g \mid h \\ h > g}} E_b[h].$$

$E_b[g]$ counts $\gcd(a, b) = g$; ordered occurrence pairs multiply by $f[b]$. **Pitfall:** zero separately; divisor subtraction adds work (e.g. $\sum_b \tau(b)^2$).

1.4 Remainder states and decimals

1.4.1 Digits as a small state

Append d: $r' = (10r + d) \bmod m$. **Use:** DP (position, r), $O(10Lm)$, only when equal states have identical legal suffixes. Add other distinguishing history to state. **Pitfall:** leading zeros; widen $10r$. Smallest reconstruction tries digits ascending with suffix feasibility, not counts alone.

1.4.2 Period and rational display

$\mathrm{ord}_m(a) \mid \phi(m)$ for $m \ge 2$, $\gcd(a, m) = 1$. **Use:** start $k = \phi(m)$; divide prime factors p while $a^{k/p} = 1 \pmod m$. Trial-factorization cost applies.

Reduce $q = b/\gcd(a, b) = 2^u 5^v r$, $\gcd(r, 10) = 1$:

$$D(a, b) = \mathrm{digits}(\lfloor a/b \rfloor) + \max(u, v) + P(r),$$

$P(1) = 0$, otherwise $P(r) = \mathrm{ord}_r(10)$. **Why:** long division removes the $2, 5$ prefix; recurring remainders then require $10^k = 1 \pmod r$. **Requires:** $a \ge 0, b > 0$; canonical decimal, no point/overbar/trailing zeros. 2025 D aggregation: gcd classes for fractions; suffix counts at $a \ge 10b, 100b, \ldots$ for integer digits. Maximum-size aggregation unbenchmarked.

DecimalPeriod.h

Description: Multiplicative order and decimal shape.

"Prelude.h", "ExactDecimals.h", "ModularArithmetic.h", "FactorSieve.h" 4d9be3, 35 lines

```
namespace finals_math {
inline i64 order_mod(
    i64 a,
    i64 m) { // m>=2, gcd(a,m)=1; trial division cost applies.
  if (m < 2 || gcd(norm(a, m), m) != 1)
    throw invalid_argument("order requires a unit");
  i64 k = phi(m);
  for (auto [p, e] : factor(k)) {
    (void)e;
    while (k % p == 0 && pow_mod(a, k / p, m) == 1) k /= p;
  }
  return k;
}
struct DecimalShape {
  int integer_digits, nonrepeating;
  i64 period;
};
// a>=0, b>=1; period computation may be costly.
inline DecimalShape decimal_shape(i64 a, i64 b) {
  if (a < 0 || b < 1)
    throw invalid_argument("nonnegative fraction");
  i64 q = b / gcd(a, b);
  int u = 0, v = 0;
  while (q % 2 == 0) {
    q /= 2;
    ++u;
  }
  while (q % 5 == 0) {
    q /= 5;
    ++v;
  }
  return {digits10(a / b), max(u, v),
          q == 1 ? 0 : order_mod(10, q)};
}
} // namespace finals_math
```

1.5 Approximation as a construction

1.5.1 Fuzzy Factorization (2025 F)

Keep ten leading digits: $X = m10^e + r$, $0 \le r < 10^e$, $m \ge 10^9$. Choose $Y = m10^e$: $|X - Y|/X < 1/m \le 10^{-9}$. **Use:** factor $m < 10^{10}$; add e to exponents of $2, 5$. At most ten digits: exact factorization. $O(|X| + \sqrt{m})$. **Requires:** positive canonical decimal text. Empty map means one; merge primes and adapt output format. Different tolerance needs a new bound.

FuzzyFactorization.h
Description: Fuzzy factorization construction.
"Prelude.h", "FactorSieve.h" 41a1d5, 21 lines

```
namespace finals_math {
// Positive decimal, no leading zero; empty factors means Y=1.
inline map<i64, int> fuzzy_factors(const string &s) {
  if (s.empty() || s[0] == '0')
    throw invalid_argument("positive canonical decimal");
  for (char c : s)
    if (c < '0' || c > '9')
      throw invalid_argument("decimal digits");
  i64 m = 0;
  int keep = int(min<size_t>(10, s.size()));
  for (int i = 0; i < keep; ++i) m = m * 10 + s[i] - '0';
  int e = int(s.size()) - keep;
  map<i64, int> f;
  for (auto [p, k] : factor(m)) f[p] = k;
  if (e) {
    f[2] += e;
    f[5] += e;
  }
  return f;
}
} // namespace finals_math
```

1.6 Combinatorics

1.6.1 Sample objects and binomial coefficients

Disjoint cases add; exactly A choices then B each multiply. Ordered distinct: $n!/(n-k)!$; subsets: $\binom{n}{k}$; ordered repetition: n^k. Stars/bars, $k \geq 1$: nonempty bins $\binom{n-1}{k-1}$ $(n \geq k)$; empty allowed $\binom{n+k-1}{k-1}$. Pascal: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ (any modulus). **Use:** factorial tables, $O(N + \log p)$ build, $O(1)$ query. **Requires:** prime p, $N < p$; primality unchecked. No Lucas/composite support.

Combinations.h
Description: Combinations under a prime modulus.
"Prelude.h", "ModularArithmetic.h" 10ccf0, 25 lines

```
namespace finals_math {
// Requires prime p and n<p; primality is unchecked.
struct PrimeCombinations {
  i64 p;
  vl fac, invfac;
  PrimeCombinations(int n, i64 prime) : p(prime) {
    if (n < 0 || p <= 1 || n >= p)
      throw invalid_argument("need 0<=n<prime");
    fac.resize(n + 1, 1);
    invfac.resize(n + 1, 1);
    for (int i = 1; i <= n; ++i)
      fac[i] = mul_mod(fac[i - 1], i, p);
    invfac[n] = pow_mod(fac[n], p - 2, p);
    for (int i = n; i > 0; --i)
      invfac[i - 1] = mul_mod(invfac[i], i, p);
  }
  i64 choose(int n, int k) const {
    if (n < 0 || n >= int(fac.size()))
      throw invalid_argument("choose range");
    if (k < 0 || k > n) return 0;
    return mul_mod(fac[n],
                   mul_mod(invfac[k], invfac[n - k], p), p);
  }
};
} // namespace finals_math
```

1.6.2 Contributions and complements

$$\sum_{\text{subarrays}} \sum a_i = \sum_i a_i(i+1)(n-i),$$

$$\sum_{i<j}(a_i + a_j) = (n-1)\sum a_i, \quad \sum_{i<j}(a_i - a_j)^2 = n\sum a_i^2 - (\sum a_i)^2.$$

At least one good = total − zero good. Bad run length L: $L(L+1)/2$ all-bad substrings. **Pitfall:** wide products; circular boundaries/all-bad case separately.

1.6.3 Inversions and repeated copies

Use: strict inversions $i < j, a_i > a_j$; compression/Fenwick, $O(n \log n)$ time, $O(n)$ space. Equal values do not count.

$$\text{inv}(S^K) = K \text{ within} + \binom{K}{2} \text{ cross}.$$

cross $= \#\{(u, v) : S_u > S_v\}$ with no within-copy index restriction (2023 I). Widen/modularize products.

Inversions.h
Description: Strict inversions by compressed Fenwick tree.
"Prelude.h", "IntegerDivision.h" 5e519b, 21 lines

```
namespace finals_math {
// Counts i<j with a[i]>a[j]; O(n log n); input unchanged.
inline i64 inversions(vl a) {
  auto vals = a;
  sort(vals.begin(), vals.end());
  vals.erase(unique(vals.begin(), vals.end()), vals.end());
  vl bit(vals.size() + 1);
  i64 answer = 0, seen = 0;
  for (i64 x : a) {
    int k = int(lower_bound(vals.begin(), vals.end(), x) -
                vals.begin()) +
            1;
    i64 le = 0;
    for (int j = k; j; j -= j & -j) le += bit[j];
    answer = narrow(i128(answer) + seen - le);
    ++seen;
    for (int j = k; j < int(bit.size()); j += j & -j) ++bit[j];
  }
  return answer;
}
} // namespace finals_math
```

1.7 Probability and expectation

$$\mathbb{E}[X] = \sum_x x \Pr(X = x), \quad \mathbb{E}[\sum I_j] = \sum \Pr(I_j = 1).$$

Linearity needs no independence; $\Pr(A \cap B) = \Pr(A)\Pr(B \mid A)$. Ordered occurrence pairs: N^2 with replacement, $N(N-1)$ without. $E = c + qE + \sum p_j E_j \Rightarrow E = (c + \sum p_j E_j)/(1-q)$, $q < 1$. **Pitfall:** modular A/B requires invertible reduced denominator; never convert rounded decimals.
$\text{Var}(X) = E[X^2] - E[X]^2$.

1.8 Proof checks

Invariant: initialize, preserve, then prove sufficiency by construction. Termination: decreasing nonnegative integer. Binary search: monotone predicate. Greedy: feasibility-preserving exchange. Tiny brute force finds counterexamples. $\sum i = n(n+1)/2$; $\sum_{i=0}^{k} r^i = (r^{k+1} - 1)/(r - 1)$, $r \neq 1$. Modular division needs cancellation/invertibility; error uses original denominator.

1.8.1 Greedy coverage: smallest missing subset sum

Positive integers sorted ascending; initially $R = 0$ and every sum in $[0, R]$ is reachable. If $x > R + 1$, answer $R + 1$; otherwise extend $R \leftarrow R + x$. After all values answer $R + 1$. Proof: old sums $[0, R]$ and sums using x, $[x, x + R]$, have no integer gap iff $x \leq R + 1$. Masters Mexico 2022 H; $O(n \log n)$; total plus one must fit.

```
long long missing(vector<long long> a) {
  sort(a.begin(), a.end());
  long long next = 1;
  for (auto x : a) {
    assert(x > 0);
    if (x > next) break;
    next += x;
  }
  return next;
}
```

1.9 Integer geometry

$u \cdot v = u_x v_x + u_y v_y$; $u \times v = u_x v_y - u_y v_x$.
orient$(A, B, C) = (B - A) \times (C - A)$: positive left, zero collinear. $2A_\triangle = |\,\text{orient}\,|$; compare squared distances. **Use:** exact predicates; segment membership = collinear + inclusive bounds. Simple polygon in boundary order: $2A = |\sum P_i \times P_{i+1}|$. **Requires:** helper coordinates $|x|, |y| \leq 10^{18}$; polygon sum needs its own bound. Compare orientation signs, not multiplied crosses; no universal floating EPS.

IntegerGeometry.h
Description: Exact integer geometry.
"Prelude.h" 45bdbc, 17 lines

```
namespace finals_math {
struct Point {
  i64 x, y;
};
// |coordinate|<=1e18 keeps these predicates within i128.
inline i128 orient(Point a, Point b, Point c) {
  return (i128(b.x) - a.x) * (i128(c.y) - a.y) -
         (i128(b.y) - a.y) * (i128(c.x) - a.x);
}
inline i128 dot(Point a, Point b) {
  return i128(a.x) * b.x + i128(a.y) * b.y;
}
inline i128 dist2(Point a, Point b) {
  i128 x = i128(a.x) - b.x, y = i128(a.y) - b.y;
  return x * x + y * y;
}
} // namespace finals_math
```